AgriVision AI

AI-Powered Smart Precision Agriculture Platform

Date: 2026-08-07 20:42

Report ID: SHR-20260807-9284

SOIL HEALTH ANALYSIS REPORT

Farmer Name: Bala

Farm Location: Patna, Bihar, India

Target Crop: General Crops

Analysis Date: 07 August 2026

Overall Soil Health Score: 75/100

Category: Good

75%

Soil Parameters & Nutrient Evaluation

Parameter	Measured Value	Optimal Range	Nutrient Status
Nitrogen (N)	45 mg/kg	30 - 60 mg/kg	**Optimal**
Phosphorus (P)	28 mg/kg	20 - 45 mg/kg	**Optimal**
Potassium (K)	40 mg/kg	30 - 60 mg/kg	**Optimal**
Soil pH	6.5	6.0 - 7.5	**Optimal (Neutral)**
Organic Carbon	0.75%	> 0.75%	**Good**
Moisture / Temp / Humidity	55% | 28°C | 62%	40-70% / 20-32°C	**Normal**

AI Agronomic & Fertilizer Recommendations

Recommended Fertilizer:	Apply 120 kg/ha Urea split into 3 doses, 60 kg/ha DAP at sowing.
Organic Manure:	Incorporate 5 tonnes/ha Vermicompost or Farmyard Manure (FYM) 2 weeks prior to sowing.
Suitable Crops:	Rice, Wheat, Tomato, Cotton, Sugarcane, Maize
Soil Improvement:	Apply Neem Cake (250 kg/ha) to enhance Nitrogen utilization and reduce soil pest risk.
Water Management:	Maintain drip irrigation schedule of 25-30 mm/week based on soil moisture levels.

Overall AI Agronomist Summary

Based on the AI soil analysis, your soil demonstrates a health score of **75/100** (Good). Nitrogen level is optimal and Phosphorus is optimal. Following the recommended NPK and organic manure regimen will optimize soil structure, microbial health, and yield potential for General Crops.

Generated by AgriVision AI Precision Agriculture Engine

Disclaimer: This report is AI-generated for agricultural decision support and should be verified with local agricultural experts before large-scale implementation.